JESNA JOSHY

AI Engineer

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OBJECTIVE

Computer Science graduate with a Diploma in AI & ML and hands-on experience in Python, Machine Learning, Computer Vision, YOLO, OpenCV, and Data Analysis. Seeking an AI/ML Apprenticeship opportunity to apply data-driven problem-solving and build scalable AI solutions.

SKILLS

Technical Skills:

- Programming Languages: Python, SQL, OOP Concepts
- Web Technologies: HTML, CSS
- Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation Techniques, Feature Engineering
- Deep Learning: TensorFlow, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN)
- Natural Language Processing (NLP): Text Preprocessing, Bag of Words (BOW), TF-IDF, Basic NLP Techniques
- Computer Vision: YOLOv8, OpenCV, Image Processing, Object Detection, Image Classification, Dataset Annotation, Roboflow
- Generative AI: Large Language Models (LLM), Prompt Engineering, AI Chatbots, Generative AI Concepts
- ML Libraries & Frameworks: Scikit-learn, TensorFlow, NLTK, OpenCV
- Data Analysis & Visualization: NumPy, Pandas, Seaborn, Excel, Matplotlib, Data cleaning, EDA
- Core Concepts: Machine Learning, Deep Learning, Data Preprocessing, EDA, Model Evaluation, OpenCV
- Databases: MySQL, NoSQL
- Tools and IDEs: Jupyter Notebook, GitHub, Tableau, Kaggle, MS Office, Roboflow

Soft Skills:

- Problem-solving, Team collaboration, Adaptability, Critical Thinking, Time Management, Analytical Thinking, Good Communication

PROFESSIONAL EXPERIENCE

Junior AI & Computer Vision Engineer

Kalgus Labs Pvt Ltd | May 2026 – Present | Ernakulam

- Developed real-time object detection systems using YOLOv8 and OpenCV for custom datasets.
- Designed and prepared custom image datasets using Roboflow.
- Trained and optimized deep learning models for classification and detection.
- Integrated Computer Vision modules with Python backend services.
- Participated in research activities related to advanced Computer Vision techniques.

Data Science Intern

Quest Innovative Solution | July 2025 – April 2026 | Ernakulam

- Processed and cleaned datasets using Python, Pandas, and NumPy.
- Performed Exploratory Data Analysis (EDA) to identify insights and patterns.
- Built Machine Learning models for prediction and classification.
- Developed visual reports and dashboards using Tableau and Matplotlib.
- Applied OpenCV for image processing tasks.
- Collaborated with team members to deliver analytical solutions.

PROJECTS

Vehicle & Number Plate Detection using YOLOv8 and PaddleOCR

2026 | Python | YOLOv8 | PaddleOCR | OpenCV

- Built an end-to-end vehicle and number plate detection pipeline, combining YOLOv8 for detection with PaddleOCR for text recognition.
- Created and annotated a custom dataset for vehicle and number plate detection.
- Trained and optimized YOLOv8 models to accurately detect vehicles and number plates in real time.
- Selected PaddleOCR for text recognition after evaluating OCR engines, as it delivered the most accurate results for reading number plates.
- Implemented position-aware OCR correction and multi-line merging logic to handle Indian number plate formats.
- Added per-track result stabilization and confidence-lock snapshots to improve reading reliability across frames.
- Integrated RTSP IP camera streaming with auto-reconnect for real-time deployment.

Restaurant Safety Compliance Detection (Gloves / No Gloves / Hairnet Detection)

2026 | Python | YOLOv8m | OpenCV | Roboflow

- Developed a real-time restaurant safety compliance detection system to identify gloves, no gloves, and hairnet usage using YOLOv8m.
- Created and annotated a custom dataset using Roboflow for training and validation.
- Trained and optimized the model to achieve above 90% detection accuracy on the prepared dataset.
- Implemented real-time object detection and monitoring to support workplace hygiene and safety compliance.
- Applied Computer Vision, Data Annotation, Model Training, and Performance Evaluation techniques.

Enhancement of Forensic Evidence Using Encryption

2025 | Python | Django | MySQL

- Developed a secure forensic evidence management system.
- Implemented encryption techniques to improve data security.
- Designed authentication and access control mechanisms.
- Enhanced integrity and confidentiality of digital evidence.

RTO Management System

2024 | Python | Django | HTML | CSS | MySQL

- Developed a web-based RTO management application.
- Implemented user registration and record management modules.
- Built secure backend operations for data storage and retrieval.
- Improved operational efficiency through digital automation.

EDUCATION

Bachelor of Science – Computer Science, 2025

Indira Gandhi College of Arts and Science, Ernakulam | Score: 68%

Advanced Diploma – Artificial Intelligence & Machine Learning, 2024 – 2025

GITH Global India Techno Hub

LANGUAGES

- Malayalam – Read / Write / Speak
- English – Read / Write / Speak